

FMUSER FUQ-4654C Encoder Modulator 1 HDMI CVBS SDI YPbPr Input 1 ASI Input DVB-S-S2 RF Output Web-Based NMS Satellite Broadcasting Headend System



Multiple Encoding interface optional

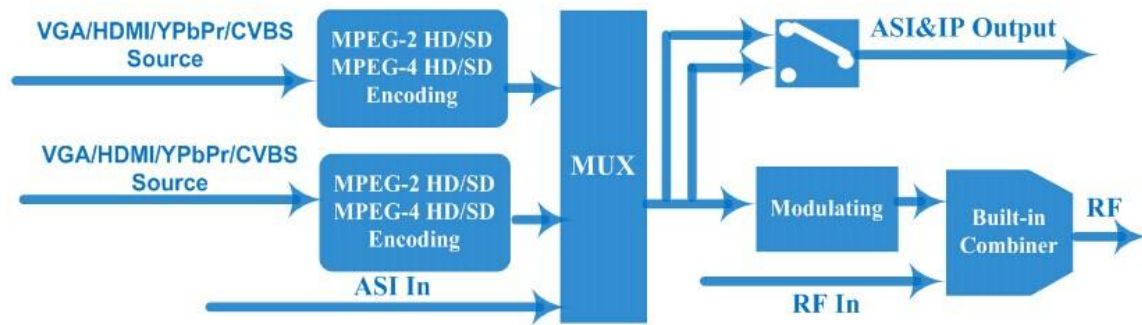
Product Overview

The FMUSER FUQ-4654C integrates encoding, multiplexing, and modulation to convert audio/video signals into DVB-S/S2 RF output. It features modular inputs for HDMI, CVBS, SDI, and YPbPr, with ASI input and outputs including ASI and IP. Controlled via web-based NMS, it is ideal for satellite broadcasting, headend systems, and flexible signal distribution.

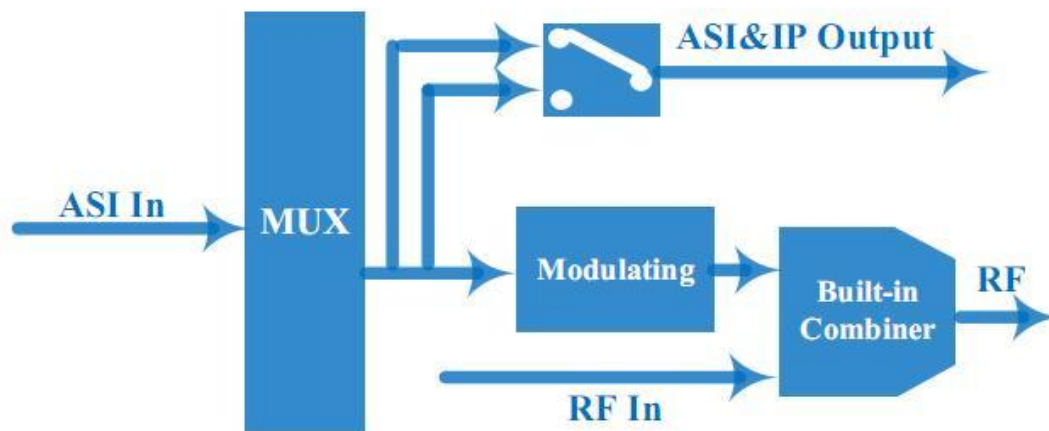
Key feature

Web-based NMS management for remote control, firmware updates, VBR/CBR rate control, and low delay encoding

Principle Chart(Encoder Modulator)



Principle Chart(Modulator)



Technical Specification

HDMI Encoding Input	Video	Input	HDMI*1(1 for backup)
HDMI Encoding Input	Video	Encoding	MPEG2; MPEG4 AVC/H.264
HDMI Encoding Input	Video	Bitrate	1-19.5Mbps
HDMI Encoding Input	Video	Resolution	1920*1080_60P, 1920*1080_50P, (-for MPEG4 AVC/H.264 only) 1920*1080_60i, 1920*1080_50i, 1280*720_60p, 1280*720_50P 720*480_60i, 720*576_50i
HDMI Encoding Input	Video	Low Delay	Normal, Mode 1, Mode 2
HDMI Encoding Input	Video	Rate Control	VBR/CBR

HDMI Encoding Input	Video	Chroma	4:2:0
HDMI Encoding Input	Video	Aspect Ratio	16:9,4:3
HDMI Encoding Input	Audio	Encoding	MPEG1 Layer II; LC-AAC; HE-AAC and Dolby Digital AC3 2.0 (Optional)
HDMI Encoding Input	Audio	Sample rate	48KHz
HDMI Encoding Input	Audio	Bitrate	64/96/128/ 192/256/320kbps

HDMI Encoding Input (Option■)	Video	Input	HDMI*1 (1 for backup)
HDMI Encoding Input (Option■)	Video	Encoding	MPEG4 AVC/H.264
HDMI Encoding Input (Option■)	Video	Bitrate	1-19.5Mbps
HDMI Encoding Input (Option■)	Video	Resolution	1920*1080_60P, 1920*1080_50P, 1920*1080_60i, 1920*1080_50i, 1280*720_60p, 1280*720_50P 720*480_60i, 720*576_50i
HDMI Encoding Input (Option■)	Video	Chroma	4:2:0
HDMI Encoding Input (Option■)	Audio	Encoding	MPEG1 Layer II
HDMI Encoding Input (Option■)	Audio	Sample rate	48KHz
HDMI Encoding Input (Option■)	Audio	Bitrate	64/96/128/ 192/256/320kbps

HDMI/YPbPr/CVBS 3-in-1 Encoding Input	Video (HDMI)	Encoding	MPEG2; MPEG4 AVC/H.264
HDMI/YPbPr/CVBS 3-in-1 Encoding Input	Video (HDMI)	Input	HDMI*1

HDMI/YPbPr/CVBS 3-in-1 Encoding Input	Video (HDMI)	Bitrate	1-19.5Mbps
HDMI/YPbPr/CVBS 3-in-1 Encoding Input	Video (HDMI)	Resolution	1920*1080_60P, 1920*1080_50P, -for MPEG4 AVC/H.264 only) 1920*1080_60i, 1920*1080_50i, 1280*720_60p, 1280*720_50P 720*480_60i, 720*576_50i
HDMI/YPbPr/CVBS 3-in-1 Encoding Input	Video (HDMI)	Low Delay	Normal, Mode 1, Mode 2
HDMI/YPbPr/CVBS 3-in-1 Encoding Input	Video (HDMI)	Rate Control	VBR/CBR
HDMI/YPbPr/CVBS 3-in-1 Encoding Input	Video (HDMI)	Chroma	4:2:0
HDMI/YPbPr/CVBS 3-in-1 Encoding Input	Video (HDMI)	Aspect Ratio	16:9,4:3
HDMI/YPbPr/CVBS 3-in-1 Encoding Input	Audio (HDMI)	Encoding	MPEG1 Layer II ,MPEG2-AAC, MPEG4-AAC and Dolby Digital AC3 2.0(Optional)
HDMI/YPbPr/CVBS 3-in-1 Encoding Input	Audio (HDMI)	Input	HDMI*1
HDMI/YPbPr/CVBS 3-in-1 Encoding Input	Audio (HDMI)	Sample rate	48KHz
HDMI/YPbPr/CVBS 3-in-1 Encoding Input	Audio (HDMI)	Bitrate	64/96/128/ 192/256/320kbps
HDMI/YPbPr/CVBS 3-in-1 Encoding Input	Video (YpbPr/ CVBS)	Encoding	MPEG2; MPEG4 AVC/H.264
HDMI/YPbPr/CVBS 3-in-1 Encoding Input	Video (YpbPr/ CVBS)	Input	YpbPr*1 / CVBS *1
HDMI/YPbPr/CVBS 3-in-1 Encoding Input	Video (YpbPr/ CVBS)	Bitrate	1-19.5Mbps

HDMI/YPbPr/CVBS 3-in-1 Encoding Input	Video (YpbPr/ CVBS)	Resolution	CVBS: 720x576_50i (PAL); 720x480_60i (NTSC) YpbPr: 1920* 1080_60i, 1920*1080_50i; 1920*1080_60p, 1920*1080_50p; 1280*720_60p, 1280*720_50P
HDMI/YPbPr/CVBS 3-in-1 Encoding Input	Video (YpbPr/ CVBS)	Low Delay	Normal, Mode 1, Mode 2
HDMI/YPbPr/CVBS 3-in-1 Encoding Input	Video (YpbPr/ CVBS)	Rate Control	VBR/CBR
HDMI/YPbPr/CVBS 3-in-1 Encoding Input	Video (YpbPr/ CVBS)	Chroma	4:2:0
HDMI/YPbPr/CVBS 3-in-1 Encoding Input	Video (YpbPr/ CVBS)	Aspect Ratio	16:9,4:3
HDMI/YPbPr/CVBS 3-in-1 Encoding Input	Audio (YpbPr/ CVBS)	Encoding	MPEG1 Layer II; MPEG2-AAC; MPEG4-AAC and Dolby Digital AC3 2.0(Optional)
HDMI/YPbPr/CVBS 3-in-1 Encoding Input	Audio (YpbPr/ CVBS)	Interface	1*Stereo/2*mono
HDMI/YPbPr/CVBS 3-in-1 Encoding Input	Audio (YpbPr/ CVBS)	Sample rate	48KHz
HDMI/YPbPr/CVBS 3-in-1 Encoding Input	Audio (YpbPr/ CVBS)	Bit rate	64/96/128/ 192/256/320kbps

2*(S-Video/YPbPr/CV BS) 3-in-1 Encoding Input	Video	Encoding	MPEG-2 & MPEG-4 AVC/H.264
2*(S-Video/YPbPr/CV BS) 3-in-1 Encoding Input	Video	Input	S-Video/YPbPr/CVBS *2
2*(S-Video/YPbPr/CV BS) 3-in-1 Encoding Input	Video	Bitrate	1-19.5Mbps

2*(S-Video/YPbPr/CV BS) 3-in-1 Encoding Input	Video	Resolution	720*480_60i, 720*576_50i
2*(S-Video/YPbPr/CV BS) 3-in-1 Encoding Input	Video	Low Delay	Normal, Mode 1, Mode 2(For option 1)
2*(S-Video/YPbPr/CV BS) 3-in-1 Encoding Input	Video	Rate Control	VBR/CBR
2*(S-Video/YPbPr/CV BS) 3-in-1 Encoding Input	Video	Chroma	4:2:0
2*(S-Video/YPbPr/CV BS) 3-in-1 Encoding Input	Video	Aspect Ratio	16:9,4:3
2*(S-Video/YPbPr/CV BS) 3-in-1 Encoding Input	Audio	Encoding	MPEG1 Layer II; LC-AAC; HE-AAC and Dolby Digital AC3 2.0(Optional)
2*(S-Video/YPbPr/CV BS) 3-in-1 Encoding Input	Audio	Sample rate	48KHz
2*(S-Video/YPbPr/CV BS) 3-in-1 Encoding Input	Audio	Bitrate	64/96/128/192/256/320kbps

SDI Encoding Input	Video	Encoding	MPEG2; MPEG4 AVC/H.264
SDI Encoding Input	Video	Input	SDI*1
SDI Encoding Input	Video	Bitrate	1-19.5Mbps
SDI Encoding Input	Video	Resolution	1920*1080_60P, 1920*1080_50P, (-for MPEG4 AVC/H.264 only) 1920*1080_60i, 1920*1080_50i, 1280*720_60p, 1280*720_50P 720*480_60i, 720*576_50i

SDI Encoding Input	Video	Low Delay	Normal, Mode 1, Mode 2
SDI Encoding Input	Video	Rate Control	VBR/CBR
SDI Encoding Input	Video	Chroma	4:2:0
SDI Encoding Input	Video	Aspect Ratio	16:9,4:3
SDI Encoding Input	Audio	Encoding	MPEG1 Layer II; LC-AAC; HE-AAC and Dolby Digital AC3 2.0 (Optional)
SDI Encoding Input	Audio	Sample rate	48KHz
SDI Encoding Input	Audio	Bitrate	64/96/128/192/256/320kbps

VGA/HDMI Encoding Input	Video (HDMI)	Encoding	MPEG2; MPEG4 AVC/H.264
VGA/HDMI Encoding Input	Video (HDMI)	Input	HDMI*1
VGA/HDMI Encoding Input	Video (HDMI)	Bitrate	1-19.5Mbps
VGA/HDMI Encoding Input	Video (HDMI)	Resolution	1920*1080_60P, 1920*1080_50P, (-for MPEG4 AVC/H.264 only) 1920*1080_60i, 1920*1080_50i, 1280*720_60p, 1280*720_50P, 720*576-50i, 720*480-60i
VGA/HDMI Encoding Input	Video (HDMI)	Low Delay	Normal, Mode 1, Mode 2
VGA/HDMI Encoding Input	Video (HDMI)	Rate Control	VBR/CBR
VGA/HDMI Encoding Input	Video (HDMI)	Chroma	4:2:0
VGA/HDMI Encoding Input	Video (HDMI)	Aspect Ratio	16:9,4:3

VGA/HDMI Encoding Input	Audio (HDMI)	Encoding	MPEG1 Layer II; LC-AAC; HE-AAC and Dolby Digital AC3 2.0 (Optional)
VGA/HDMI Encoding Input	Audio (HDMI)	Sample rate	48KHz
VGA/HDMI Encoding Input	Audio (HDMI)	Bitrate	64/96/128/192/256/320kbps
VGA/HDMI Encoding Input	Video (VGA)	Encoding	MPEG2; MPEG4 AVC/H.264
VGA/HDMI Encoding Input	Video (VGA)	Input	VGA(SVGA/XGA/UXGA/SXGA)
VGA/HDMI Encoding Input	Video (VGA)	Bitrate	1-19.5Mbps
VGA/HDMI Encoding Input	Video (VGA)	Resolution	1920*1080_60P, 1280*720_60p
VGA/HDMI Encoding Input	Video (VGA)	Low Delay	Normal, Mode 1, Mode 2
VGA/HDMI Encoding Input	Video (VGA)	Rate Control	VBR/CBR
VGA/HDMI Encoding Input	Video (VGA)	Chroma	4:2:0
VGA/HDMI Encoding Input	Video (VGA)	Aspect Ratio	16:9,4:3
VGA/HDMI Encoding Input	Audio (VGA)	Encoding	MPEG1 Layer II; LC-AAC; HE-AAC and Dolby Digital AC3 2.0 (Optional)
VGA/HDMI Encoding Input	Audio (VGA)	Sample rate	48KHz
VGA/HDMI Encoding Input	Audio (VGA)	Bit rate	64/96/128/192/256/320kbps

Modulator Section	DVB-S/S2	Constellation	DVB-S: QPSK DVB-S2: QPSK, 8PSK
Modulator Section	DVB-S/S2	FEC	1/2,2/3,3/4,5/6,7/8 (DVB-S QPSK)

Modulator Section	DVB-S/S2	FEC	1/2,3/5,2/3,3/4,4/5,5/6,8/9,9/10 (DVB-S2 QPSK)
Modulator Section	DVB-S/S2	FEC	3/5,2/3,3/4,5/6,8/9,9/10 (DVB-S2 8PSK)
Modulator Section	DVB-S/S2	RF output	950.00-1050.00 MHz, 1Khz step
Modulator Section	DVB-S/S2	Symbol rate	0.05-40.0MSPS (DVB-S and DVB-S2 QPSK)
Modulator Section	DVB-S/S2	Symbol rate	0.05-26.0MSPS (DVB-S2 8PSK)
Modulator Section	DVB-S/S2	Roll Off	0.35 (DVB-S)
Modulator Section	DVB-S/S2	Roll Off	0.35, 0.25, 0.2 (DVB-S2)

General	System	Local interface	LCD + control buttons
General	System	Remote management	Web NMS
General	System	Stream Out	2 ASI out (BNC type); IP (MPTS/SPTS) out (100M)
General	System	NMS interface	RJ45, 100M
General	System	Language	English
General	Physical Specification	Power supply	AC 100V~240V
General	Physical Specification	Dimensions	482*300*44mm
General	Physical Specification	Weight	4.5 kg
General	Physical Specification	Operation temperature	0~45℃

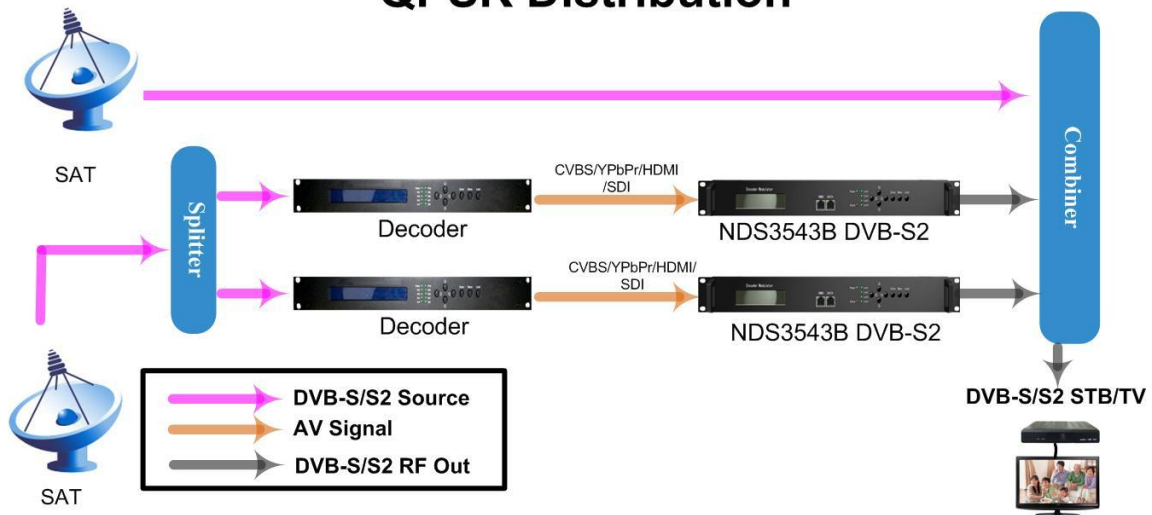
Key Features

	FU-4654B	FU-4654C	FU-4513E
Encode In	2 encoding slots with different encoding boards available	1 encoding slot with different encoding boards available	N/A
ASI In	1 ASI input for re-mux	1 ASI input for pass-through	4 ASI inputs with 3 for backup

IP In	N/A	N/A	1 IP Input over UDP (100M)
RF Range	950 – 1050 MHz	950 -2150 MHz	950 -2150 MHz
Symbol Rate	QPSK: 0.05-40.0Msps 8PSK: 0.05-26.0Msps	0.05-20.0Msps	32APSK: 0.5~40Msps 16APSK/8PSK/QPSK: 0.5~45 Msps
Constellation	QPSK, 8PSK	QPSK, 8PSK	QPSK, 8PSK, 16APSK, 32APSK
Output Level	-30 dBm ~-10.0 dBm	-10.0 dBm ~-41.5 dBm	-10.0 dBm~-41.5 dBm
BISS Scramble	N/A	Support	Support
IP output	1 IP over UDP (100M)	N/A	N/A
10MHz Ref Clock	N/A	Support	Support
24V RF Out	N/A	Support	Support
RF In for Mix	Support	N/A	N/A
SFN TS Transmission	N/A	N/A	Support

Application Case

QPSK Distribution



About Us

FMUSER is a company specializing in broadcasting and television transmission systems and integrating hotel IPTV systems. The company was established in 2009 and has served over 180 countries worldwide. More than 20,000 television stations, radio stations, and hotel clients from different.

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